

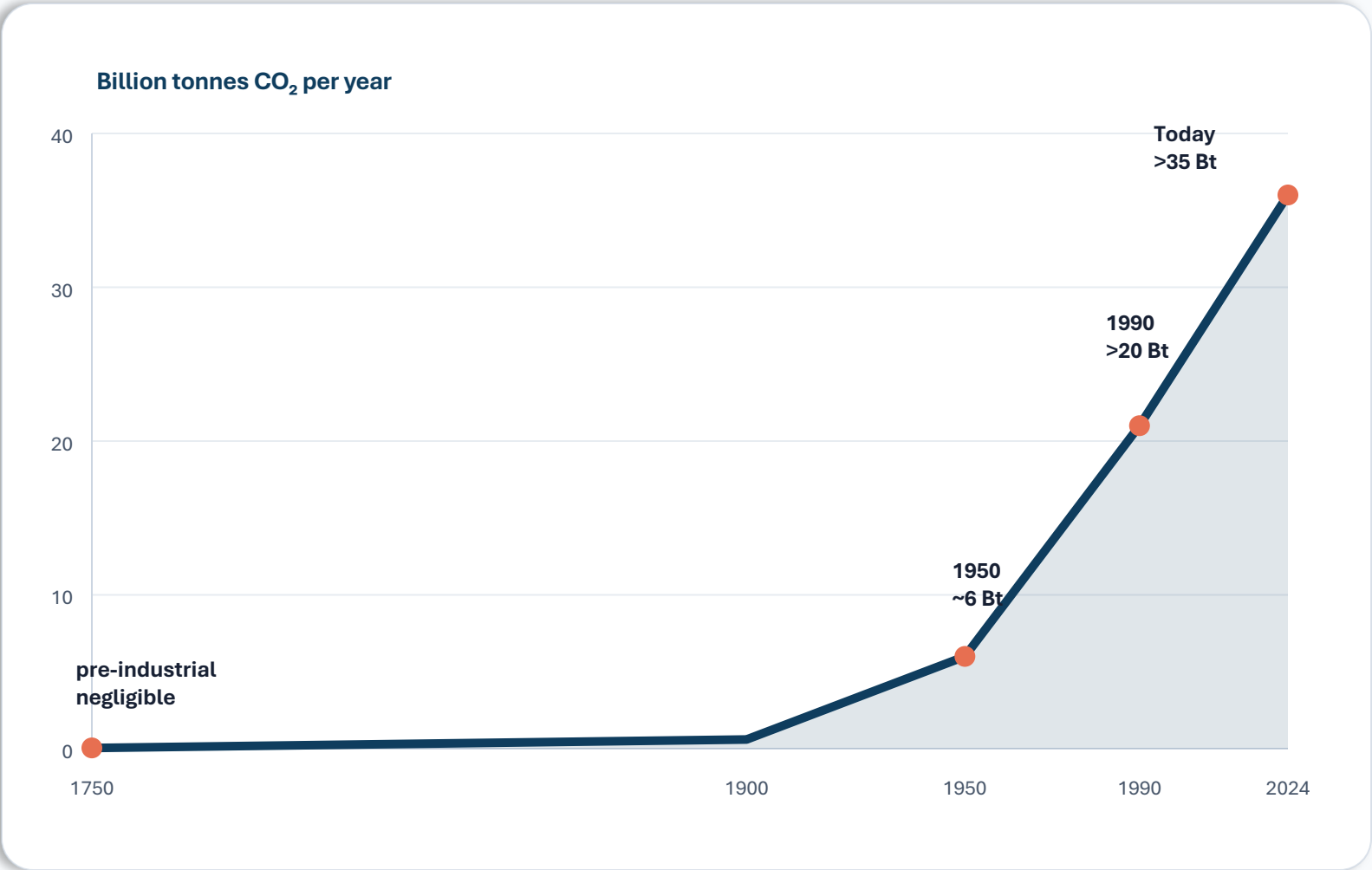
Global CO₂ Emissions: Who Emits, How Much, and What's Changing

Scope: CO₂ from fossil fuels and industry, 1750–2024

Prepared for policy analysts, climate negotiators, and sustainability researchers

Global CO₂ Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today

Emissions rose from negligible pre-industrial levels to >35 Bt annually; growth has slowed, but the peak has not arrived.



6 Bt

1950 annual emissions

>20 Bt

By 1990, nearly quadrupled

>35 Bt

Today, with no peak yet

Interpretation: the climate policy challenge is not only historic growth, but the absence of a clear global turning point.

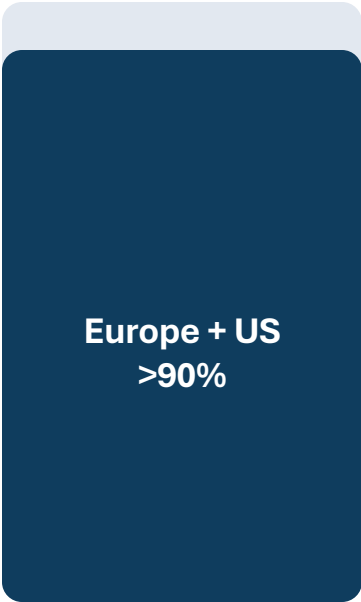
Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today

The center of annual emissions shifted from the early industrial economies to Asia, led by China.

Stacked-share visual (values shown only where provided in source brief)

Key reading

- Europe and the United States dominated early industrial emissions.
- Their combined share remained above 85% in 1950.
- Today they are less than one-third, while Asia produces roughly half.



1900



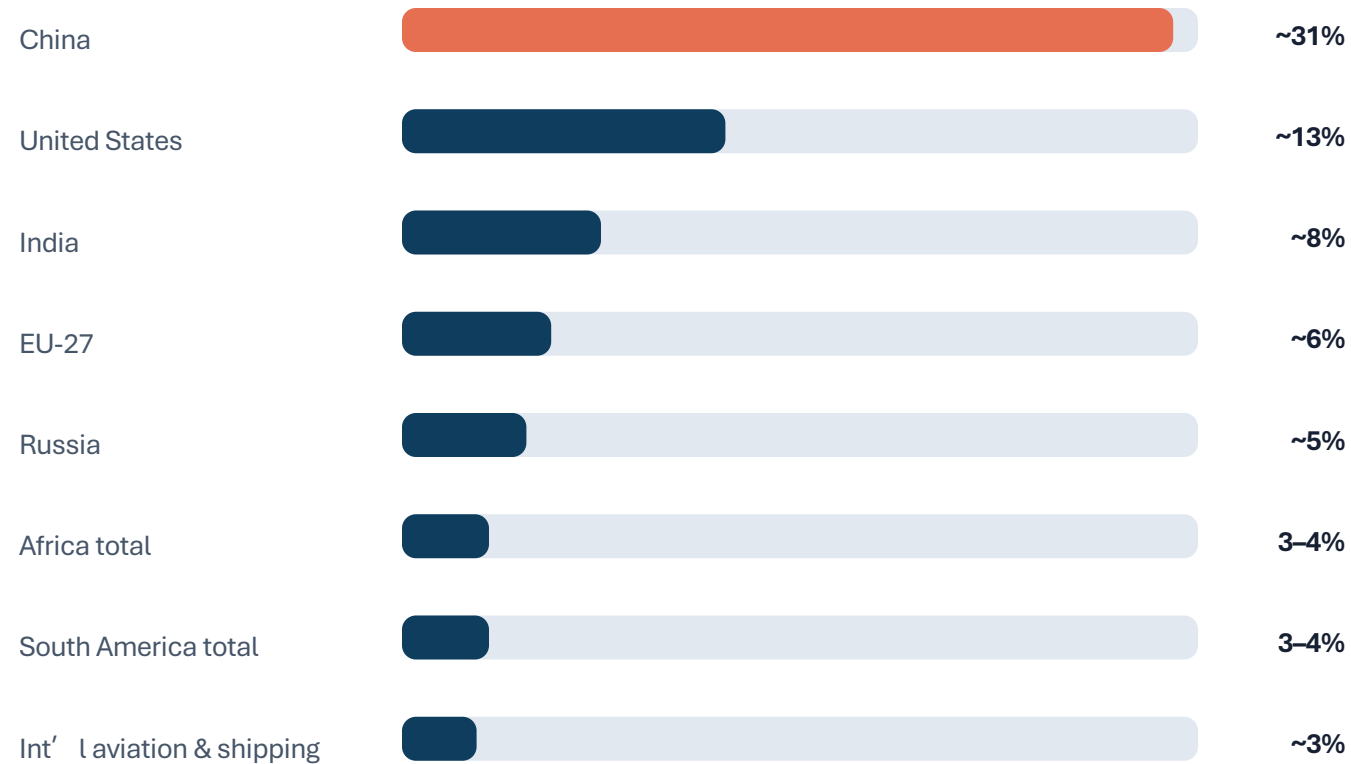
1950



Today

China Now Emits More Than One-Quarter of Global CO₂

In 2024, China accounts for ~31% of global CO₂, larger than the US, India, and EU-27 combined shares shown here.



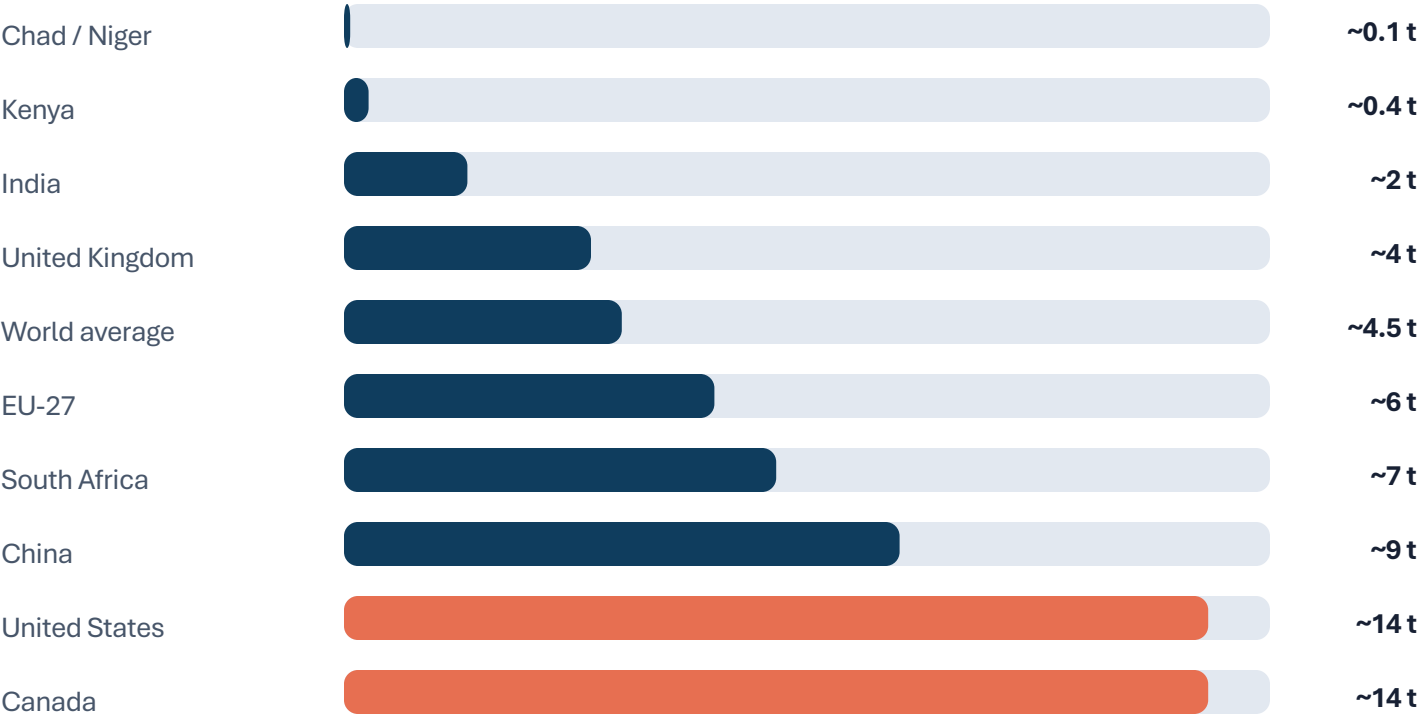
What this reframes

- 1 The largest annual emitter is now China, not the early industrial economies.
- 2 Africa and South America each contribute only 3–4% — comparable to international aviation and shipping.
- 3 Country shares and per capita responsibility point to different policy narratives.

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada

A horizontal bar chart makes the responsibility gap visible: roughly 150x separates the lowest emitters from high-emitting large countries.

Tonnes CO₂ per person, approx. 2024



150x

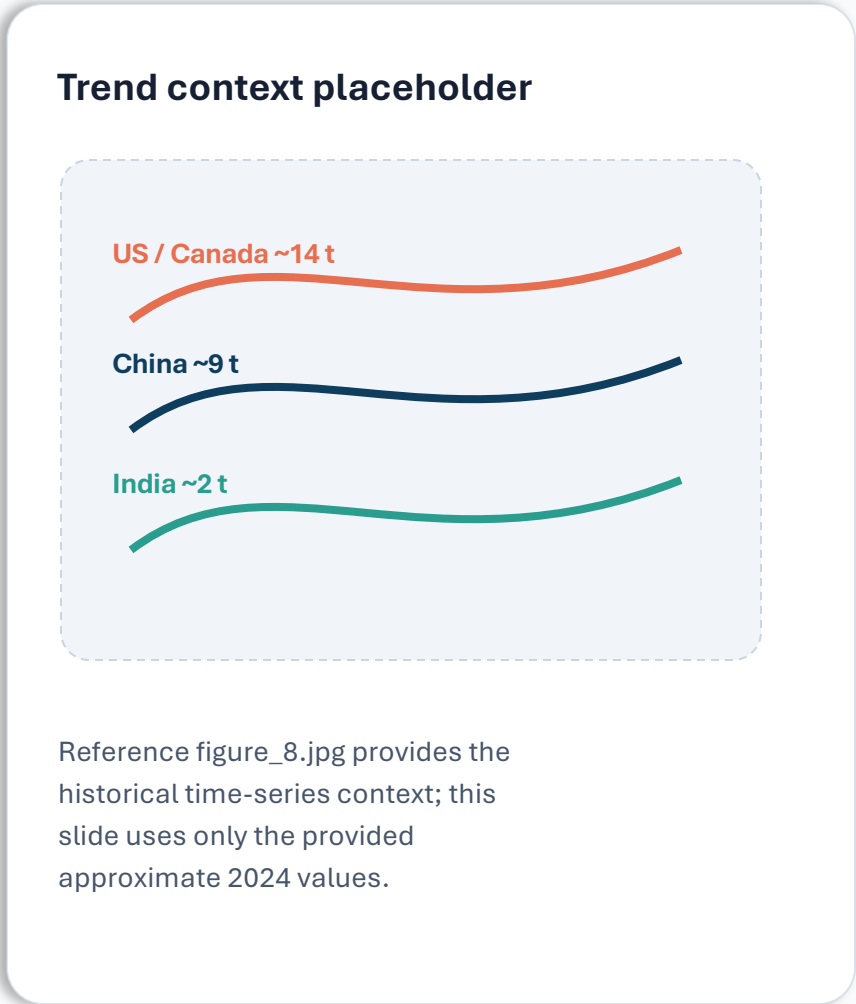
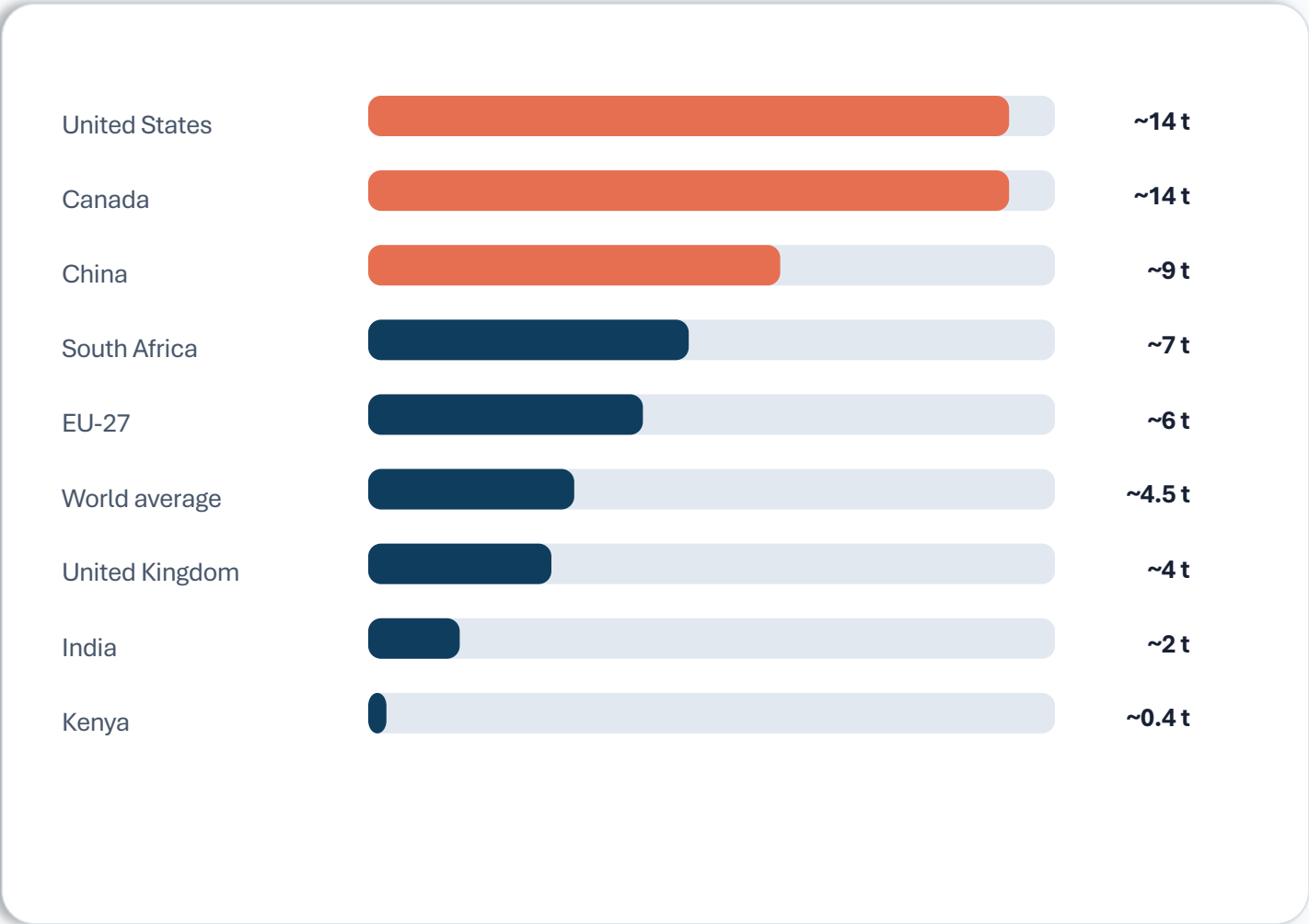
gap between the lowest emitters and high-emitting large countries

The average American or Australian emits in under two days what the average person in Mali or Niger emits in an entire year.

Per capita measures reframe “who emits” around people, not just national totals.

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

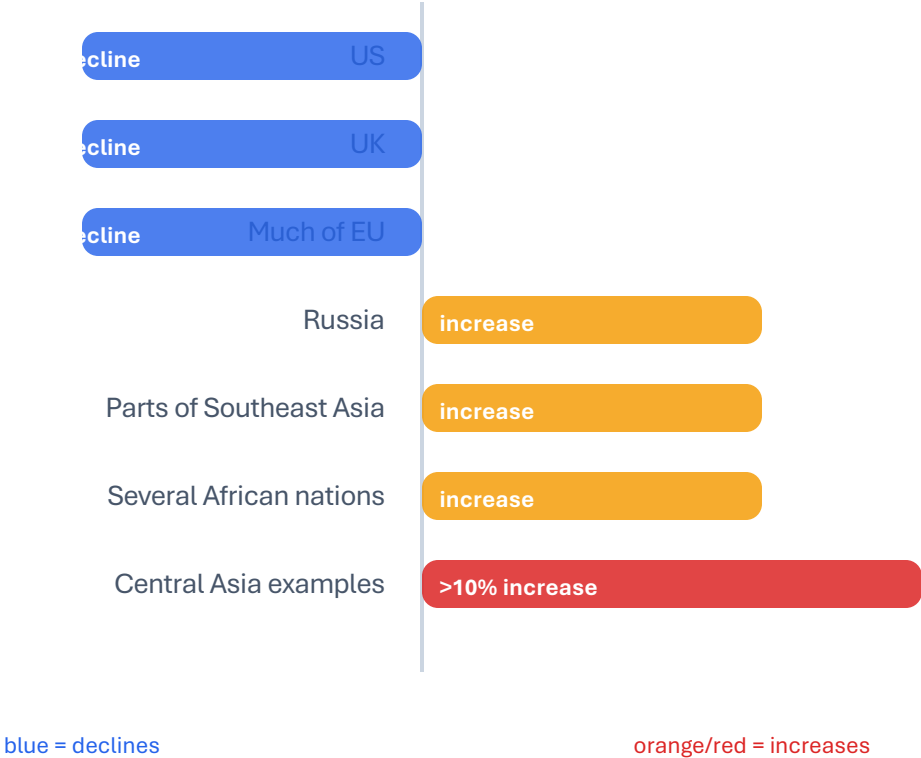
China is the largest total emitter, but its per-person emissions remain well below the US and Canada; India remains very low.



2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

The 2024 map is mixed: declines in many Western nations, increases in Russia, parts of Southeast Asia, and several African nations.

Diverging bar placeholder: annual % change in CO₂ emissions, 2024



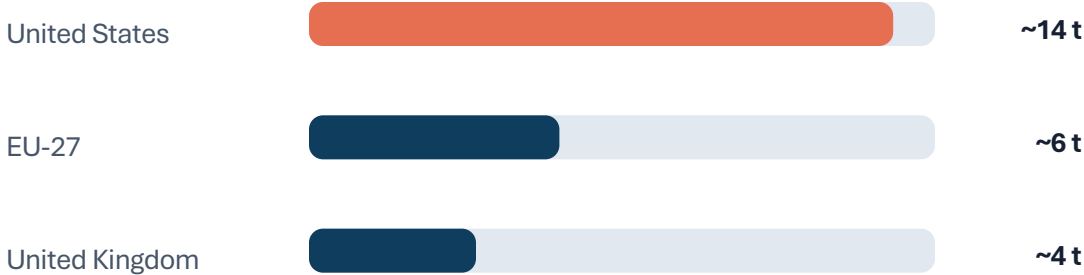
Policy implication

- Decarbonization progress is visible in many developed economies.
- Global totals can still rise if growth elsewhere offsets these reductions.
- The source map flags some Central Asian countries with increases exceeding 10%.

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US

Among wealthy nations, prosperity alone does not determine emissions; electricity choices shape per-person outcomes.

Available per capita figures from source brief



France, the UK, and Portugal have much lower per capita emissions than Germany, the Netherlands, or the US.

Driver logic

Similar prosperity levels

wealthy countries can still differ sharply

Energy mix choices

higher nuclear + renewable electricity share

Lower per capita CO₂

France, UK, Portugal lower than Germany, Netherlands, US

Key Takeaways: Responsibility Is Shared but Unequal

1

Global emissions exceed 35 Bt/yr and have not peaked.

2

China is the largest annual emitter, but per capita emissions remain below the US.

3

Historical responsibility is dominated by Europe and the US.

4

Per capita gaps of 150x persist between richest and poorest nations.

5

Energy policy and technology choices can decouple prosperity from emissions.